NEET UG PHYSICS TUTORIAL

Mechanical Properties of Solids

Comprehensive Theory, Reference Data, Worked Examples, and NEET UG Practice Questions

1. Learning Objectives & Chapter Overview

In ideal rigid-body mechanics, a solid object is treated as a hard body with a fixed shape and size. In reality, however, no solid body is perfectly rigid. Even an appreciably rigid steel bar can be stretched, compressed, or bent when a sufficiently large external force is applied to it. This tutorial explores how solids deform under external forces and how internal restoring forces govern their mechanical behaviour.

Key Learning Objectives for NEET UG:

- Distinguish between **elasticity** (tendency to regain original shape and size upon removal of force) and **plasticity** (permanent deformation without gross tendency to recover, as in putty or mud).
- Define and calculate the three fundamental types of **stress** (tensile/compressive, shearing, and hydraulic) and their corresponding **strains** (longitudinal, shearing, and volume).
- Apply **Hooke's law** and interpret every region of a metallic **stress-strain curve**, distinguishing between ductile Materials, brittle materials, and elastomers (such as rubber and the elastic tissue of the aorta).
- Solve quantitative problems involving **Young's modulus (Y)**, **Shear modulus (G)**, **Bulk modulus (B)**, **compressibility (k)**, **Poisson's ratio**, and **elastic potential energy (U)** stored in a stretched wire.
- Understand real-world engineering applications: crane rope thickness design, sagging and buckling of loaded beams (and the use of I-shaped cross-sections), column design, and the maximum possible height of mountains on Earth.

2. Core Concepts and Explanations

2.1 Elasticity and Plasticity

A solid has a definite shape and size. To change (or deform) the shape or size of a body, a force is required:

- **Elasticity:** If you stretch a helical spring by gently pulling its ends, its length increases slightly. When you release the ends, it regains its original size and shape. The property of a body by virtue of which it tends to regain its original size and shape when the applied force is removed is known as **elasticity**, and the deformation caused is known as **elastic deformation**.
- **Plasticity:** If you apply force to a lump of putty or mud, they have no gross tendency to regain their previous shape and get permanently deformed. Such substances are called **plastic** and this property is called **plasticity**. Putty and mud are close to ideal plastics.

2.2 Stress and Strain

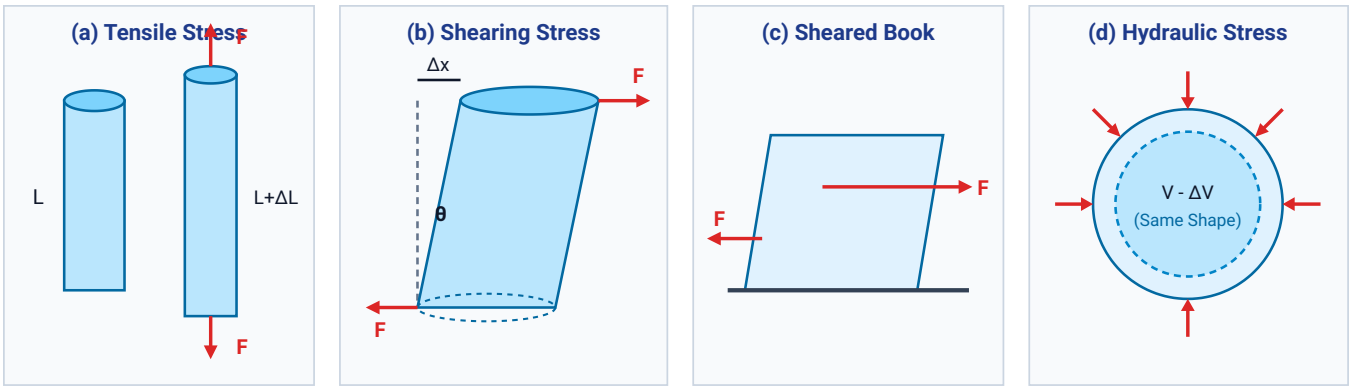
When forces are applied on a body in such a manner that the body is still in static equilibrium, it is deformed to a small or large extent depending upon the nature of the material and the magnitude of the deforming force. When a body is subjected to a deforming force, a **restoring force** is developed inside the body. This restoring force is equal in magnitude but opposite in direction to the applied deforming force.

Definition of Stress: The restoring force per unit area is known as **stress**. If F is the force applied normal to the cross-section and A is the area of cross-section of the body:

$$\text{Magnitude of stress} = \frac{F}{A}$$

- **SI Unit of Stress:** $\mathrm{N\,m^{-2}}$ or pascal (Pa)
- **Dimensional Formula of Stress:** $[\mathrm{M\,L^{-1}\,T^{-2}}]$

There are three distinct ways in which a solid may change its dimensions when an external force acts on it:



Tutorial Diagram 1: (a) A cylindrical body under tensile stress elongates by ΔL. (b) Shearing stress on a cylinder deforming it by angle θ. (c) A book subjected to shearing stress by pushing horizontally. (d) A solid body under uniform normal stress (hydraulic compression) undergoes volumetric strain $\Delta V/V$ with no change in geometrical shape.

1. Longitudinal Stress (Tensile or Compressive) and Longitudinal Strain

When a cylinder is stretched by two equal forces applied normal to its cross-sectional area, the restoring force per unit area is called **tensile stress**. If the cylinder is compressed under the action of applied forces, the restoring force per unit area is known as **compressive stress**. Tensile or compressive stress is collectively termed **longitudinal stress**.

In both cases, there is a change in the length of the cylinder. The ratio of the change in length (ΔL) to the original length (L) of the body is known as **longitudinal strain**:

$$\text{Longitudinal strain} = \frac{\Delta L}{L}$$

2. Tangential (or Shearing) Stress and Shearing Strain

If two equal and opposite deforming forces are applied parallel to the cross-sectional area of a cylinder (or when a book is pressed with the hand and pushed horizontally), there is a relative displacement between the opposite faces of the body. The restoring force per unit area developed due to the applied tangential force is known as **tangential stress** or **shearing stress**.

The strain produced is known as **shearing strain**, defined as the ratio of the relative displacement of the faces (Δx) to the length of the cylinder (L):

$$\text{Shearing strain} = \frac{\Delta x}{L} = \tan\theta$$

where θ is the angular displacement of the cylinder from the vertical (its original position). Usually θ is very small, so $\tan\theta$ is nearly equal to the angle θ in radians (for example, if $\theta = 10^\circ$, there is only a 1% difference between θ and $\tan\theta$):

$$\text{Shearing strain} = \tan\theta \approx \theta$$

3. Hydraulic Stress and Volume Strain

When a solid sphere is placed in a fluid under high pressure, it is compressed uniformly on all sides. The force applied by the fluid acts in a perpendicular direction at each point of the surface, and the body is said to be under **hydraulic compression**. This leads to a decrease in its volume without any change in its geometrical shape.

The body develops internal restoring forces equal and opposite to the forces applied by the fluid (the body restores its original shape and size when taken out from the fluid). The internal restoring force per unit area in this case is known as **hydraulic stress**, which in magnitude is equal to the hydraulic pressure p (applied force per unit area).

The strain produced by a hydraulic pressure is called **volume strain** (or volumetric strain), defined as the ratio of change in volume (ΔV) to the original volume (V):

$$\text{Volume strain} = \frac{\Delta V}{V}$$

Note: Since strain is a ratio of a change in dimension to the original dimension, **strain has no units and no dimensional formula**.

2.3 Hooke's Law

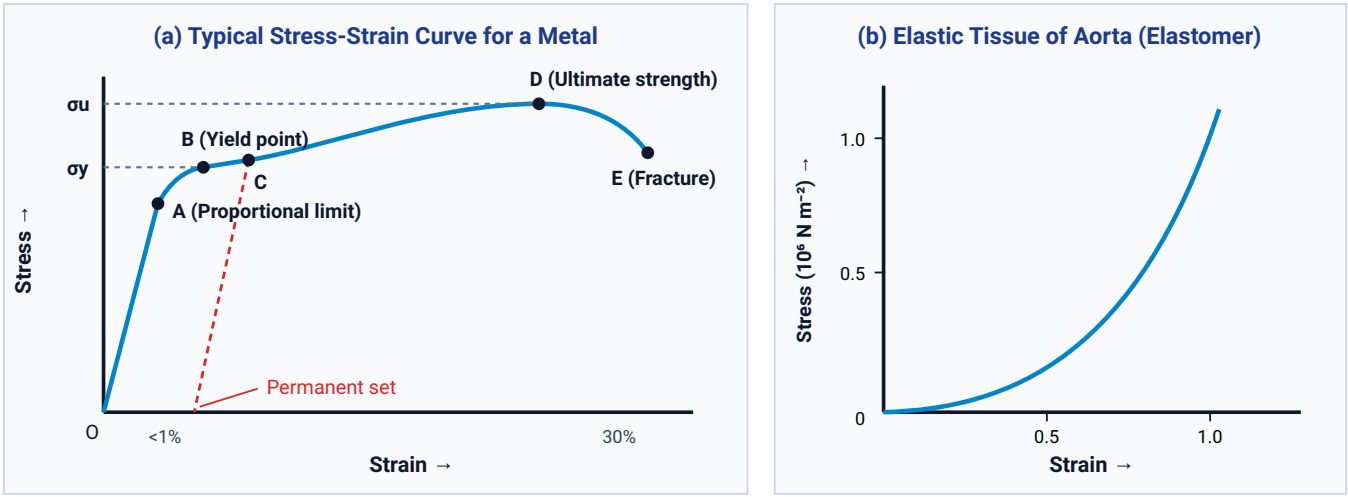
For small deformations, the stress and strain are directly proportional to each other. This is known as **Hooke's law**:

$$\text{stress} \propto \text{strain} \implies \text{stress} = k \times \text{strain}$$

where k is the proportionality constant known as the **modulus of elasticity**. Hooke's law is an empirical law and is found to be valid for most materials, though some materials (like elastomers) do not exhibit this linear relationship.

2.4 Stress-Strain Curve

The relation between stress and strain for a given material under tensile stress can be found experimentally by stretching a test cylinder or wire with an applied force that is gradually increased in steps, recording the fractional change in length (strain) and the force per unit area (stress).



Tutorial Diagram 2: (a) A typical stress-strain curve for a metal showing key regions and points O, A, B, C, D, and E. (b) Stress-strain curve for the elastic tissue of the aorta (the large vessel carrying blood from the heart), an elastomer.

Detailed Analysis of the Metallic Stress-Strain Curve:

1. **Proportional Region (O to A):** The curve is linear. In this region, **Hooke's law is obeyed**. The body regains its original dimensions completely when the applied force is removed, behaving as an elastic body. Point A is the **proportional limit**.
2. **Non-Linear Elastic Region (A to B):** Stress and strain are no longer proportional. Nevertheless, the body still returns to its original dimension when the load is removed. Point B is known as the **yield point** (also called the **elastic limit**), and the corresponding stress is the **yield strength (σ_y)** of the material.
3. **Plastic Deformation and Permanent Set (B to D):** If the load is increased beyond the yield point B, the stress developed exceeds the yield strength and the strain increases rapidly even for a small change in stress. If the load is removed at some point C between B and D, the body does not regain its original dimension; even when the stress becomes zero, the residual strain is non-zero. The material is said to have a **permanent set**, and the deformation is called **plastic deformation**.
4. **Ultimate Tensile Strength and Fracture (D to E):** Point D on the graph represents the **ultimate tensile strength (σ_u)** of the material. Beyond point D, additional strain is produced even by a reduced applied force, and **fracture** occurs at point E.

Material Classification from Stress-Strain Curves:

- **Brittle Materials:** If the ultimate strength point D and fracture point E are **close to each other** (e.g., glass), the material fractures soon after passing the elastic limit and is said to be **brittle**.
- **Ductile Materials:** If points D and E are **far apart** (e.g., copper, steel, brass), the material undergoes substantial plastic deformation before fracturing and is said to be **ductile**.
- **Elastomers:** Substances like rubber and the elastic tissue of the aorta can be pulled to cause very large strains (e.g., strain of 1.0 or several times original length) and still return to their original shape. As seen in the aorta curve, although the elastic region is very large, **elastomers do not obey Hooke's law over most of the region**, and there is **no well-defined plastic region**.

3. Elastic Moduli, Reference Data & Important Equations

Within the proportional region (OA), the ratio of stress to strain is a constant characteristic of the material called the **modulus of elasticity**.

3.1 Young's Modulus (Y)

Experimental observations show that for a given material, the magnitude of the strain produced is the same whether the stress is tensile or compressive. The ratio of tensile (or compressive) stress (σ) to the longitudinal strain (ε) is defined as **Young's modulus (Y)**:

$$Y = \frac{\sigma}{\varepsilon} = \frac{F/A}{\Delta L/L} = \frac{F \times L}{A \times \Delta L}$$

The SI unit of Young's modulus is $\mathrm{N\,m^{-2}}$ or Pa. Metals have large Young's moduli and therefore require a large force to produce a small change in length. For instance, to increase the length of a thin wire of $0.1\ \mathrm{cm^2}$ cross-sectional area by 0.1%, a force of $2000\ \mathrm{N}$ is required for steel, compared to $690\ \mathrm{N}$ for aluminium, $900\ \mathrm{N}$ for brass, and $1100\ \mathrm{N}$ for copper. Thus, **steel is more elastic than copper, brass, and aluminium**, which is why steel is preferred in heavy-duty machines and structural designs.

Substance	Density ρ ($\mathrm{kg\,m^{-3}}$)	Young's Modulus Y ($10^9\ \mathrm{N\,m^{-2}}$)	Ultimate Strength σ_u ($10^6\ \mathrm{N\,m^{-2}}$)	Yield Strength σ_y ($10^6\ \mathrm{N\,m^{-2}}$)
Aluminium	2710	70	110	95
Copper	8890	110	400	200
Iron (wrought)	7800–7900	190	330	170
Steel	7860	200	400	250
Glass*	2190	65	50	—
Concrete	2320	30	40	—
Wood*	525	13	50	—
Bone*	1900	9.4	170	—
Polystyrene	1050	3	48	—

*Substances tested under compression.

3.2 Shear Modulus or Modulus of Rigidity (G)

The ratio of shearing stress (σ_s) to the corresponding shearing strain is called the **shear modulus (G)** or **modulus of rigidity**:

$$G = \frac{\text{shearing stress } (\sigma_s)}{\text{shearing strain}} = \frac{F/A}{\Delta x/L} = \frac{F \times L}{A \times \Delta x} = \frac{F}{A \times \theta}$$

$$\sigma_s = G \times \theta$$

The SI unit of shear modulus is $\mathrm{N\,m^{-2}}$ or Pa. Shear modulus is generally less than Young's modulus; for most materials:

$$G \approx \frac{Y}{3}$$

Material	Shear Modulus G ($10^9\ \mathrm{N\,m^{-2}}$ or GPa)	Material	Shear Modulus G ($10^9\ \mathrm{N\,m^{-2}}$ or GPa)
Aluminium	25	Lead	5.6
Brass	36	Nickel	77
Copper	42	Steel	84
Glass	23	Tungsten	150
Iron	70	Wood	10

3.3 Bulk Modulus (B) and Compressibility (k)

When a body is submerged in a fluid, it undergoes hydraulic stress equal in magnitude to the hydraulic pressure p, leading to a volume strain $\Delta V/V$. The ratio of hydraulic stress to the corresponding hydraulic (volume) strain is called the **bulk modulus (B)**:

$$B = -\frac{p}{\Delta V/V}$$

The negative sign indicates that with an increase in pressure ($p > 0$), a decrease in volume ($\Delta V < 0$) occurs, ensuring that B is always positive for a system in equilibrium. Its SI unit is $\mathrm{N\,m^{-2}}$ or Pa.

The reciprocal of the bulk modulus is called **compressibility (k)**, defined as the fractional change in volume per unit increase in pressure:

$$k = \frac{1}{B} = -\frac{1}{\Delta p}\left(\frac{\Delta V}{V}\right)$$

Physical Insight across States of Matter: Bulk moduli for solids are much larger than for liquids, which in turn are much larger than for gases (air). Gases are about a million times more compressible than solids! Solids are the least compressible due to tight coupling between neighbouring atoms; molecules in liquids are bound less strongly; and molecules in gases are very poorly coupled to their neighbours.

Category	Material	Bulk Modulus B ($10^9\,\mathrm{N\,m^{-2}}$ or GPa)
Solids	Aluminium	72
	Brass	61
	Copper	140
	Glass	37
	Iron	100
	Nickel	260
	Steel	160
Liquids	Water	2.2
	Ethanol	0.9
	Carbon disulphide	1.56
	Glycerine	4.76
	Mercury	25
Gases	Air (at STP)	1.0×10^{-4}

3.4 Summary Comparison of Stress, Strain, and Elastic Moduli

Type of Stress	Description of Stress	Strain	Change in Shape?	Change in Volume?	Elastic Modulus	State of Matter
Tensile or Compressive ($\sigma = F/A$)	Two equal and opposite forces perpendicular to opposite faces	Longitudinal strain ($\Delta L/L$)	Yes	No	Young's modulus $Y = \frac{FL}{A\Delta L}$	Solid
Shearing ($\sigma_s = F/A$)	Two equal and opposite forces parallel to opposite surfaces (net force and net torque vanish)	Pure shear ($\theta = \Delta x/L$)	Yes	No	Shear modulus $G = \frac{F}{A\theta}$	Solid
Hydraulic ($p = F/A$)	Forces perpendicular everywhere to the surface; pressure same everywhere	Volume strain ($\Delta V/V$)	No	Yes	Bulk modulus $B = -\frac{p}{\Delta V/V}$	Solid, Liquid, and Gas

3.5 Poisson's Ratio

When a wire is stretched along its length, its length increases while its cross-sectional diameter decreases. The strain perpendicular to the applied force is called **lateral strain**. Simon Poisson pointed out that within the elastic limit, lateral strain is directly proportional to the longitudinal strain. The ratio of the lateral strain to the longitudinal strain in a stretched wire is called **Poisson's ratio**:

$$\text{Poisson's ratio} = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{\Delta d/d}{\Delta L/L} = \left(\frac{\Delta d}{\Delta L}\right)\left(\frac{L}{d}\right)$$

where d is the original diameter, Δd is the contraction of the diameter under stress, L is the original length, and ΔL is the elongation. Being a ratio of two strains, Poisson's ratio is a pure dimensionless number with no units. For steels, its value is between 0.28 and 0.30, and for aluminium alloys it is about 0.33.

3.6 Elastic Potential Energy in a Stretched Wire

When a wire is put under tensile stress, work is done against inter-atomic forces and is stored in the wire as **elastic potential energy**. For a wire of original length L, cross-sectional area A, and Young's modulus Y, the restoring force at an instantaneous elongation l is:

$$F = \frac{YAl}{L}$$

The work done dW for a further infinitesimal elongation dl is $dW = F\,dl = \frac{YAl}{L}\,dl$. Integrating from $l = 0$ to $l = l$ gives the total work done W (stored as elastic potential energy U):

$$W = U = \int_0^l \frac{YAl}{L}\,dl = \frac{1}{2}\frac{YAl^2}{L} = \frac{1}{2}Fl$$

Rewriting in terms of stress, strain, and the volume of the wire ($V = AL$):

$$U = \frac{1}{2} \times Y \times \left(\frac{l}{L}\right)^2 \times (AL) = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume of the wire}$$

Therefore, the **elastic potential energy per unit volume** (u) (also called energy density) of the stretched wire is:

$$u = \frac{U}{AL} = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}Y\varepsilon^2 = \frac{\sigma^2}{2Y}$$

4. Applications of Elastic Behaviour of Materials

4.1 Designing Metal Ropes for Cranes

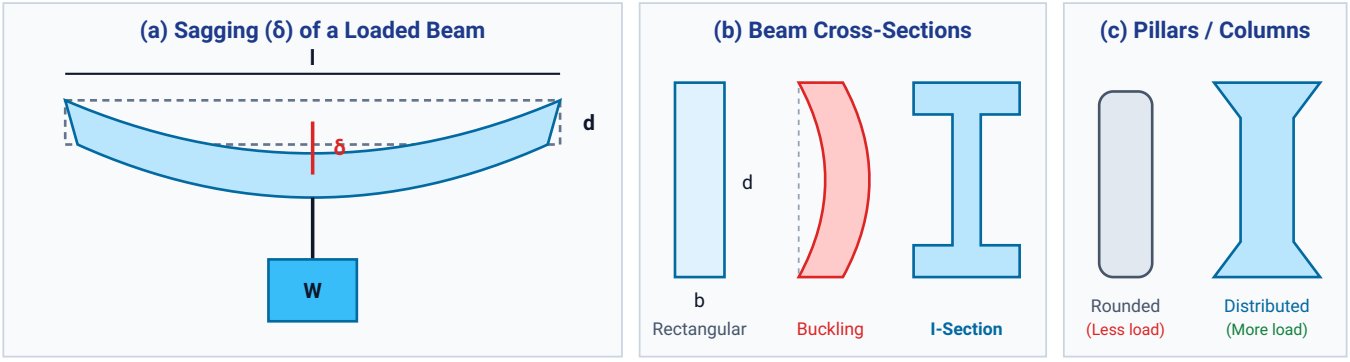
Cranes used for lifting heavy loads use thick metal ropes. To prevent permanent deformation, the stress in the rope must not exceed the yield strength (σ_y) of the steel. For a lifting capacity of mass $M = 10\,\text{tonnes} = 10^4\,\text{kg}$ and mild steel with yield strength $\sigma_y = 300 \times 10^6\,\text{N m}^{-2}$, the minimum cross-sectional area A required is:

$$A \geq \frac{W}{\sigma_y} = \frac{Mg}{\sigma_y} = \frac{10^4\,\text{kg} \times 9.8\,\text{m s}^{-2}}{300 \times 10^6\,\text{N m}^{-2}} \approx 3.3 \times 10^{-4}\,\text{m}^2$$

For a circular cross-section ($\pi r^2 = A$), this corresponds to a minimum radius of about $1\,\text{cm}$. In practice, a large safety margin (a factor of about 10 in the load) is provided, so a rope of radius about $3\,\text{cm}$ is recommended. Because a single solid rod of $3\,\text{cm}$ radius would be rigid, crane ropes are always made of numerous thin wires braided together (like pigtails) for ease in manufacture, flexibility, and strength.

4.2 Bending of Beams, Buckling, and I-Shaped Girders

In bridges and buildings, beams and columns must support heavy loads without bending excessively or breaking. Consider a bar of length l, breadth b, and depth d supported near its ends and loaded at the centre by a weight W.



Tutorial Diagram 3: (a) A beam of length l and depth d loaded at the centre sags by δ. (b) Rectangular bar, lateral buckling of a deep thin bar, and the optimal I-shaped cross-section. (c) Pillars with rounded ends support less load than pillars with distributed ends.

Formula for Central Sagging (δ) of a Symmetrically Supported Beam:

$$\delta = \frac{Wl^3}{4bd^3Y}$$

- To reduce bending (δ) for a given load W, we must use a material with a **large Young's modulus** Y and keep the span length l as small as possible ($\delta \propto l^3$).

- For a given cross-sectional area, **increasing the depth d rather than the breadth b is far more effective** in reducing bending because $\delta \propto d^{-3}$ whereas $\delta \propto b^{-1}$.
- However, if the bar is made very deep and thin, it may bend sideways under off-centre moving traffic loads—a phenomenon called **buckling**.
- **Why I-Shaped Beams (Girders) Are Used:** An **I-shaped cross-section** provides a large load-bearing surface (top and bottom flanges) to prevent buckling and enough depth d to prevent sagging. It also removes unnecessary material from the central web, significantly reducing the weight of the beam without sacrificing strength and hence reducing manufacturing cost.

4.3 Design of Pillars and Columns

In buildings and bridges, a pillar with rounded ends supports substantially less load than a pillar with a distributed (flared) shape at the ends, because distributed ends spread the compressive force over a larger contact area and resist buckling.

4.4 Maximum Height of a Mountain on Earth

Why is the maximum height of a mountain on Earth approximately $10\,\text{km}$? The base of a mountain is not under uniform hydraulic compression because its sides are free. The weight of the overlying rock creates a vertical stress $h\rho g$ at the bottom of a mountain of height h, which produces an approximate **shearing stress component equal to** $h\rho g$. If this stress exceeds the critical shearing stress (elastic limit) of the rock, the rock at the base flows plastically.

For a typical rock, the critical shearing stress (elastic limit) is $\sigma_c = 30 \times 10^7\,\text{N}\,\text{m}^{-2}$ and its density is $\rho = 3 \times 10^3\,\text{kg}\,\text{m}^{-3}$. Taking $g = 10\,\text{m}\,\text{s}^{-2}$:

$$h\rho g = 30 \times 10^7\,\text{N}\,\text{m}^{-2} \implies h = \frac{30 \times 10^7\,\text{N}\,\text{m}^{-2}}{3 \times 10^3\,\text{kg}\,\text{m}^{-3} \times 10\,\text{m}\,\text{s}^{-2}} = 10^4\,\text{m} = 10\,\text{km}$$

This theoretical limit ($\sim 10\,\text{km}$) is just slightly greater than the height of Mount Everest!

5. Complete Worked Examples

The following worked examples cover every standard problem archetype from the syllabus.

Worked Example 1: Stress, Elongation, and Strain in a Structural Rod

Problem: A structural steel rod has a radius of $10\,\text{mm}$ and a length of $1.0\,\text{m}$. A $100\,\text{kN}$ force stretches it along its length. Calculate (a) stress, (b) elongation, and (c) strain on the rod. Young's modulus of structural steel is $2.0 \times 10^{11}\,\text{N}\,\text{m}^{-2}$.

Solution:

Given: $r = 10\,\text{mm} = 10^{-2}\,\text{m}$, $L = 1.0\,\text{m}$, $F = 100\,\text{kN} = 100 \times 10^3\,\text{N}$, and $Y = 2.0 \times 10^{11}\,\text{N}\,\text{m}^{-2}$.

(a) Stress:

$$\text{Stress} = \frac{F}{A} = \frac{F}{\pi r^2} = \frac{100 \times 10^3\,\text{N}}{3.14 \times (10^{-2}\,\text{m})^2} = 3.18 \times 10^8\,\text{N}\,\text{m}^{-2}$$

(b) Elongation (ΔL):

$$\Delta L = \frac{(F/A)L}{Y} = \frac{(3.18 \times 10^8\,\text{N}\,\text{m}^{-2}) \times (1.0\,\text{m})}{2.0 \times 10^{11}\,\text{N}\,\text{m}^{-2}} = 1.59 \times 10^{-3}\,\text{m} = 1.59\,\text{mm}$$

(c) Strain:

$$\text{Strain} = \frac{\Delta L}{L} = \frac{1.59 \times 10^{-3}\,\text{m}}{1.0\,\text{m}} = 1.59 \times 10^{-3}\ (\text{or } 0.16\%)$$

Worked Example 2: Composite Wires Connected in Series

Problem: A copper wire of length $2.2\,\mathrm{m}$ and a steel wire of length $1.6\,\mathrm{m}$, both of diameter $3.0\,\mathrm{mm}$, are connected end to end. When stretched by a load, the net elongation is found to be $0.70\,\mathrm{mm}$. Obtain the load applied. (Given: $Y_c = 1.1 \times 10^{11}\,\mathrm{N\,m^{-2}}$ and $Y_s = 2.0 \times 10^{11}\,\mathrm{N\,m^{-2}}$.)

Solution:

Because both wires are connected in series and have the same diameter $d = 3.0\,\mathrm{mm}$ (radius $r = 1.5 \times 10^{-3}\,\mathrm{m}$), they experience the exact same tension W and have the same cross-sectional area $A = \pi r^2$. Therefore, the tensile stress W/A is identical in both wires:

$$\frac{W}{A} = Y_c\left(\frac{\Delta L_c}{L_c}\right) = Y_s\left(\frac{\Delta L_s}{L_s}\right)$$

$$\frac{\Delta L_c}{\Delta L_s} = \left(\frac{Y_s}{Y_c}\right) \times \left(\frac{L_c}{L_s}\right) = \left(\frac{2.0 \times 10^{11}}{1.1 \times 10^{11}}\right) \times \left(\frac{2.2}{1.6}\right) = 2.5$$

We are given the total elongation:

$$\Delta L_c + \Delta L_s = 0.70\,\mathrm{mm} = 7.0 \times 10^{-4}\,\mathrm{m}$$

Substituting $\Delta L_c = 2.5\,\Delta L_s$ into the sum gives $3.5\,\Delta L_s = 7.0 \times 10^{-4}\,\mathrm{m}$, so:

$$\Delta L_s = 2.0 \times 10^{-4}\,\mathrm{m} \quad \text{and} \quad \Delta L_c = 5.0 \times 10^{-4}\,\mathrm{m}$$

Now, the applied load W is:

$$W = \frac{A \times Y_c \times \Delta L_c}{L_c} = \frac{3.14 \times (1.5 \times 10^{-3}\,\mathrm{m})^2 \times (1.1 \times 10^{11}\,\mathrm{N\,m^{-2}}) \times (5.0 \times 10^{-4}\,\mathrm{m})}{2.2\,\mathrm{m}} \approx 1.8 \times 10^2\,\mathrm{N}$$

Worked Example 3: Biomechanical Compression of Thighbones (Femurs)

Problem: In a human pyramid in a circus, the entire weight of the balanced group is supported by the legs of a performer who is lying on his back. The combined mass of all the persons performing the act, and the tables, plaques, etc. involved is $280\,\mathrm{kg}$. The mass of the performer lying on his back at the bottom of the pyramid is $60\,\mathrm{kg}$. Each thighbone (femur) of this performer has a length of $50\,\mathrm{cm}$ and an effective radius of $2.0\,\mathrm{cm}$. Determine the amount by which each thighbone gets compressed under the extra load. (Young's modulus for bone $Y = 9.4 \times 10^9\,\mathrm{N\,m^{-2}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$.)

Solution:

Extra mass supported by the legs of the bottom performer $= 280\,\mathrm{kg} - 60\,\mathrm{kg} = 220\,\mathrm{kg}$.

Total weight of this supported mass $= 220 \times 9.8\,\mathrm{N} = 2156\,\mathrm{N}$.

Weight supported by *each* of the two thighbones: $F = \frac{2156\,\mathrm{N}}{2} = 1078\,\mathrm{N}$.

Cross-sectional area of each thighbone: $A = \pi r^2 = 3.14 \times (2.0 \times 10^{-2}\,\mathrm{m})^2 = 1.26 \times 10^{-3}\,\mathrm{m^2}$.

Compression ΔL in each thighbone of length $L = 0.5\,\mathrm{m}$:

$$\Delta L = \frac{F \times L}{Y \times A} = \frac{1078\,\mathrm{N} \times 0.5\,\mathrm{m}}{(9.4 \times 10^9\,\mathrm{N\,m^{-2}}) \times (1.26 \times 10^{-3}\,\mathrm{m^2})} = 4.55 \times 10^{-5}\,\mathrm{m} = 4.55 \times 10^{-3}\,\mathrm{cm}$$

The fractional decrease in length is $\Delta L/L = 0.000091$ or 0.0091%.

Worked Example 4: Shear Deflection of a Lead Slab

Problem: A square lead slab of side $50\,\text{cm}$ and thickness $10\,\text{cm}$ is subjected to a shearing force (on its narrow face) of $9.0 \times 10^4\,\text{N}$. The lower edge is riveted to the floor. How much will the upper edge be displaced? (Shear modulus of lead $G = 5.6 \times 10^9\,\text{N m}^{-2}$.)

Solution:

The shearing force $F = 9.0 \times 10^4\,\text{N}$ is applied parallel to the narrow face of dimensions $50\,\text{cm} \times 10\,\text{cm}$. Thus, the area parallel to the force is:

$$A = 0.5\,\text{m} \times 0.1\,\text{m} = 0.05\,\text{m}^2$$

The shearing stress applied is:

$$\sigma_s = \frac{F}{A} = \frac{9.0 \times 10^4\,\text{N}}{0.05\,\text{m}^2} = 1.80 \times 10^6\,\text{N m}^{-2}$$

Since $G = \frac{\text{Stress} \times L}{\Delta x}$, where $L = 0.5\,\text{m}$ is the vertical height of the slab, the displacement Δx of the upper edge is:

$$\Delta x = \frac{\text{Stress} \times L}{G} = \frac{1.80 \times 10^6\,\text{N m}^{-2} \times 0.5\,\text{m}}{5.6 \times 10^9\,\text{N m}^{-2}} = 1.6 \times 10^{-4}\,\text{m} = 0.16\,\text{mm}$$

Worked Example 5: Fractional Compression of Water at the Bottom of the Ocean

Problem: The average depth of the Indian Ocean is about $3000\,\text{m}$. Calculate the fractional compression, $\Delta V/V$, of water at the bottom of the ocean, given that the bulk modulus of water is $2.2 \times 10^9\,\text{N m}^{-2}$. (Take $g = 10\,\text{m s}^{-2}$ and density of water $\rho = 1000\,\text{kg m}^{-3}$.)

Solution:

The excess hydraulic pressure exerted by a $3000\,\text{m}$ column of water at the bottom is:

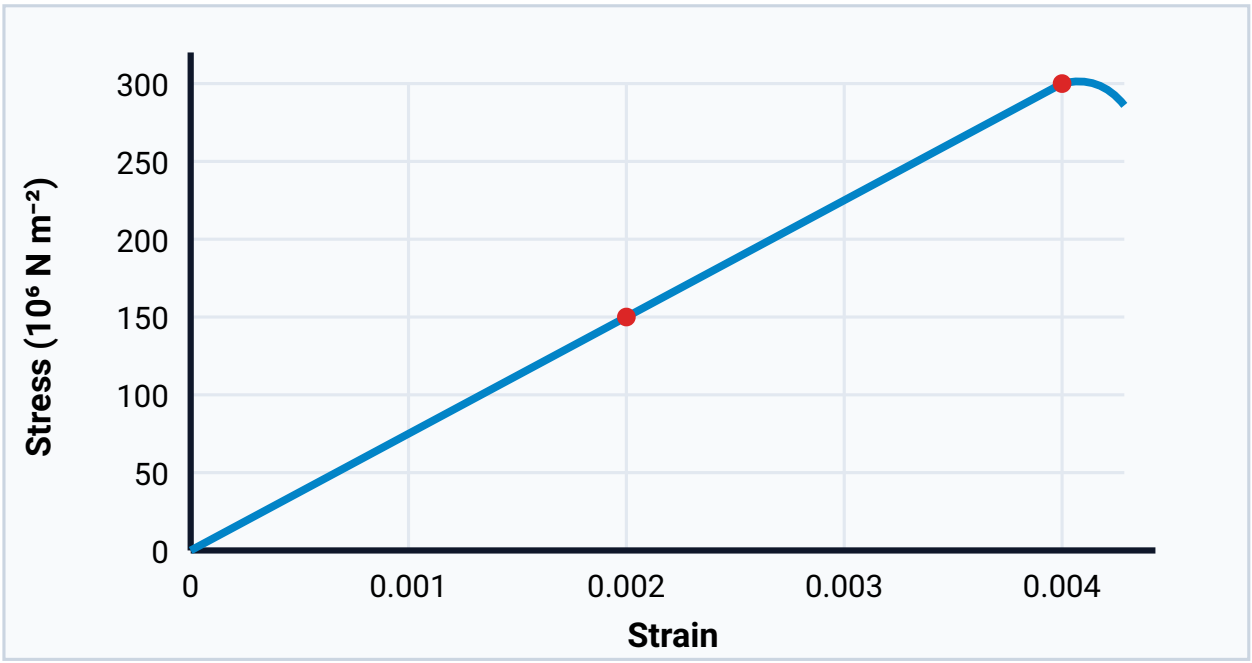
$$p = h\rho g = 3000\,\text{m} \times 1000\,\text{kg m}^{-3} \times 10\,\text{m s}^{-2} = 3 \times 10^7\,\text{N m}^{-2}$$

Therefore, the magnitude of fractional compression $\Delta V/V$ is:

$$\frac{\Delta V}{V} = \frac{p}{B} = \frac{3 \times 10^7\,\text{N m}^{-2}}{2.2 \times 10^9\,\text{N m}^{-2}} = 1.36 \times 10^{-2}\ (\text{or } 1.36\%)$$

Worked Example 6: Reading Young's Modulus and Yield Strength from a Graph

Problem: Examine the stress-strain graph for a given material shown below, and determine (a) Young's modulus and (b) the approximate yield strength for this material.



Solution:

(a) Young's Modulus (Y): In the linear region, the graph passes through a stress of $150 \times 10^6\,\mathrm{N\,m^{-2}}$ at a strain of 0.002 (and $300 \times 10^6\,\mathrm{N\,m^{-2}}$ at a strain of 0.004). The slope of the linear region gives Young's modulus:

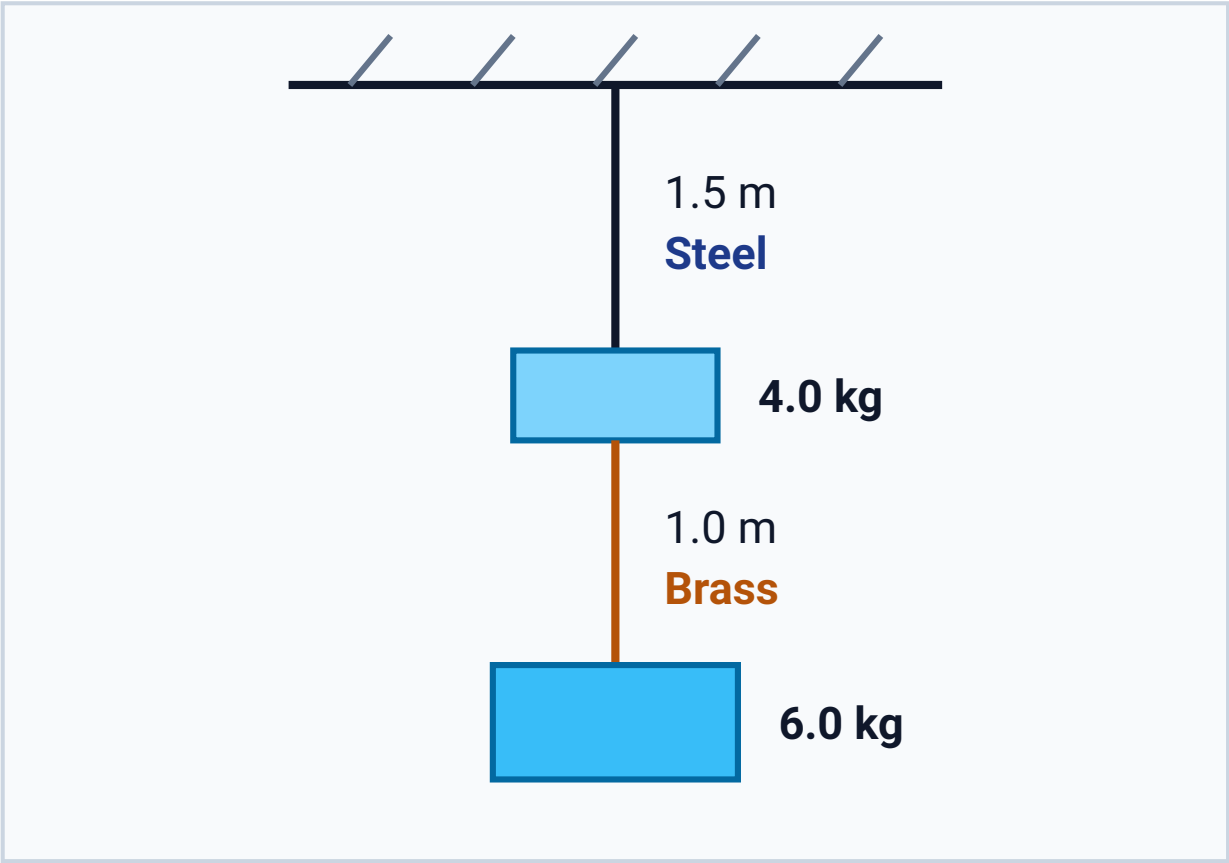
$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{150 \times 10^6\,\mathrm{N\,m^{-2}}}{0.002} = 7.5 \times 10^{10}\,\mathrm{N\,m^{-2}} = 75\,\mathrm{GPa}$$

(b) Approximate Yield Strength (σ_y): The elastic limit (yield point) where the curve bends away from linearity occurs at the peak of the linear segment, corresponding to a stress of:

$$\sigma_y \approx 300 \times 10^6\,\mathrm{N\,m^{-2}} = 3 \times 10^8\,\mathrm{N\,m^{-2}}$$

Worked Example 7: Two Suspended Wires with Intermediate and Lower Loads

Problem: Two wires of diameter $0.25\ \text{cm}$, one made of steel and the other made of brass, are loaded as shown in the diagram below. The unloaded length of the upper steel wire is $1.5\ \text{m}$ and that of the lower brass wire is $1.0\ \text{m}$. Compute the elongations of the steel and the brass wires. (Given: $Y_{\text{steel}} = 2.0 \times 10^{11}\ \text{Pa}$, $Y_{\text{brass}} = 0.91 \times 10^{11}\ \text{Pa}$, $g = 9.8\ \text{m s}^{-2}$.)



Solution:

Radius of each wire: $r = \frac{0.25\ \text{cm}}{2} = 0.125 \times 10^{-2}\ \text{m}$. Cross-sectional area: $A = \pi r^2 = 3.14 \times (1.25 \times 10^{-3}\ \text{m})^2 \approx 4.91 \times 10^{-6}\ \text{m}^2$.

1. Upper Steel Wire: The steel wire supports *both* the $4.0\ \text{kg}$ mass and the $6.0\ \text{kg}$ mass. Thus, the total tension in the steel wire is:

$$F_s = (4.0 + 6.0)\ \text{kg} \times 9.8\ \text{m s}^{-2} = 98\ \text{N}$$

Elongation of the steel wire ($L_s = 1.5\ \text{m}$):

$$\Delta L_s = \frac{F_s L_s}{A Y_{\text{steel}}} = \frac{98\ \text{N} \times 1.5\ \text{m}}{(4.91 \times 10^{-6}\ \text{m}^2) \times (2.0 \times 10^{11}\ \text{Pa})} \approx 1.5 \times 10^{-4}\ \text{m}$$

2. Lower Brass Wire: The brass wire supports *only* the lower $6.0\ \text{kg}$ mass. Thus, the tension in the brass wire is:

$$F_b = 6.0\ \text{kg} \times 9.8\ \text{m s}^{-2} = 58.8\ \text{N}$$

Elongation of the brass wire ($L_b = 1.0\ \text{m}$):

$$\Delta L_b = \frac{F_b L_b}{A Y_{\text{brass}}} = \frac{58.8\ \text{N} \times 1.0\ \text{m}}{(4.91 \times 10^{-6}\ \text{m}^2) \times (0.91 \times 10^{11}\ \text{Pa})} \approx 1.3 \times 10^{-4}\ \text{m}$$

Worked Example 8: Wire Whirled in a Vertical Circle & Density Change Under Pressure

Problem Part A: A $14.5\,\text{kg}$ mass, fastened to the end of a steel wire of unstretched length $1.0\,\text{m}$, is whirled in a vertical circle with an angular velocity of $2\,\text{rev/s}$ at the bottom of the circle. The cross-sectional area of the wire is $0.065\,\text{cm}^2$. Calculate the elongation of the wire when the mass is at the lowest point of its path. ($Y_{\text{steel}} = 2.0 \times 10^{11}\,\text{N m}^{-2}$, $g = 9.8\,\text{m s}^{-2}$.)

Problem Part B: What is the density of water at a depth where pressure is $80.0\,\text{atm}$, given that its density at the surface is $1.03 \times 10^3\,\text{kg m}^{-3}$? ($B_{\text{water}} = 2.2 \times 10^9\,\text{Pa}$, $1\,\text{atm} = 1.013 \times 10^5\,\text{Pa}$.)

Solution to Part A:

Angular velocity $\omega = 2\,\text{rev/s} = 2 \times 2\pi\,\text{rad/s} = 4\pi\,\text{rad/s} \approx 12.56\,\text{rad/s}$.

At the lowest point of the vertical circle, the tension T in the wire provides the centripetal force and balances gravity:

$$T = mg + mr\omega^2 = 14.5 \times 9.8 + 14.5 \times 1.0 \times (4\pi)^2 = 142.1\,\text{N} + 2289.8\,\text{N} = 2431.9\,\text{N}$$

With $A = 0.065\,\text{cm}^2 = 0.065 \times 10^{-4}\,\text{m}^2$, the elongation ΔL is:

$$\Delta L = \frac{TL}{AY} = \frac{2431.9\,\text{N} \times 1.0\,\text{m}}{(6.5 \times 10^{-6}\,\text{m}^2) \times (2.0 \times 10^{11}\,\text{N m}^{-2})} \approx 1.87 \times 10^{-3}\,\text{m} = 1.87\,\text{mm}$$

Solution to Part B:

Excess pressure $p \approx 80.0\,\text{atm} = 80.0 \times 1.013 \times 10^5\,\text{Pa} = 8.104 \times 10^6\,\text{Pa}$. Fractional decrease in volume:

$$\frac{\Delta V}{V} = \frac{p}{B} = \frac{8.104 \times 10^6\,\text{Pa}}{2.2 \times 10^9\,\text{Pa}} = 3.68 \times 10^{-3}$$

For a fixed mass M of water, $V' = V - \Delta V = V\left(1 - \frac{\Delta V}{V}\right)$. Therefore, the new density ρ' is:

$$\rho' = \frac{\rho}{1 - (\Delta V/V)} \approx \rho\left(1 + \frac{\Delta V}{V}\right) = (1.03 \times 10^3\,\text{kg m}^{-3}) \times (1 + 3.68 \times 10^{-3}) \approx 1.034 \times 10^3\,\text{kg m}^{-3}$$

6. NEET UG Practice Questions with Complete Worked Solutions

Note: The following original practice questions are specifically designed to match the NEET UG examination standard, covering single-choice conceptual, graphical, assertion-reason, multi-statement, and numerical problems.

Practice Question 1 (Conceptual — Tension and Stress)

A uniform metal wire of cross-sectional area A is suspended vertically from a rigid ceiling, and a block of weight W is attached to its lower end. In static equilibrium, neglecting the mass of the wire, the ceiling exerts an upward force of magnitude W on the upper end of the wire. What are the tension and the tensile stress at any horizontal cross-section of the wire?

(1) Tension is $2W$ and tensile stress is $\frac{2W}{A}$

(2) Tension is W and tensile stress is $\frac{W}{A}$

(3) Tension is 0 and tensile stress is 0

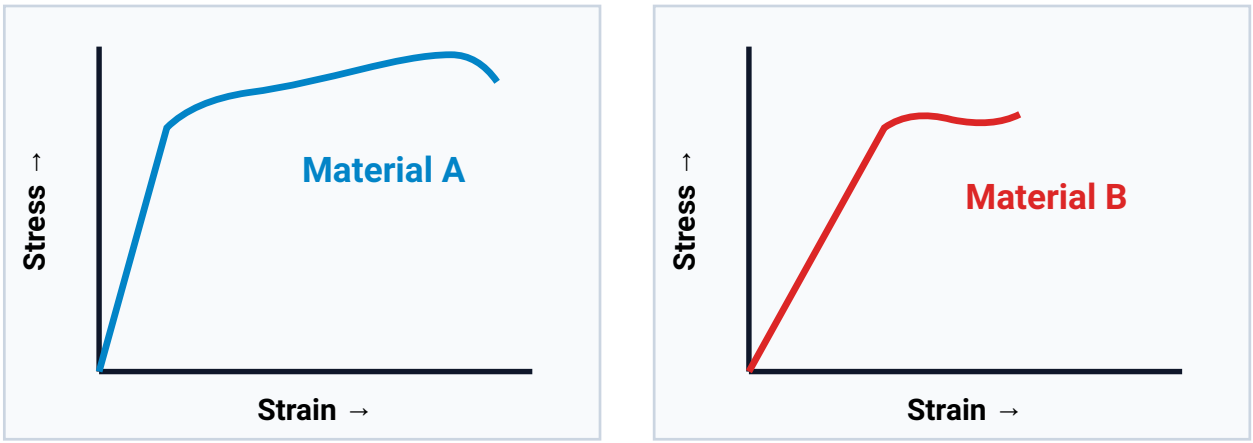
(4) Tension is $\frac{W}{2}$ and tensile stress is $\frac{W}{2A}$

Correct Answer: (2)

Worked Solution: When a wire is suspended from a ceiling and stretched by a weight W at its lower end, the ceiling exerts an equal and opposite upward force W to maintain static equilibrium. If we imagine cutting the wire at any cross-section A and consider the free-body diagram of the lower portion, the internal restoring force (tension) balancing the weight is simply W, **not** $2W$. Therefore, the tensile stress at any cross-section is $\frac{W}{A}$.

Practice Question 2 (Graphical Analysis — Stress-Strain Curves)

The stress-strain graphs for two metallic materials A and B, drawn to the exact same scale, are shown below.



Based on these graphs, which of the following statements is correct?

(1) Material A has a greater Young's modulus and is stronger than Material B.

(2) Material B has a greater Young's modulus and is stronger than Material A.

(3) Material A has a greater Young's modulus, whereas Material B is stronger.

(4) Material B has a greater Young's modulus, whereas Material A is stronger.

Correct Answer: (1)

Worked Solution:

1. **Young's Modulus (Y):** Young's modulus is equal to the slope of the linear (proportional) part of the stress-strain graph ($Y = \frac{\text{Stress}}{\text{Strain}}$). Since the linear portion of Material A has a steeper slope than that of Material B, **Material A has the greater Young's modulus**.
2. **Strength of a Material:** The strength of a material is determined by the height of the ultimate tensile stress peak (σ_u), i.e., the maximum stress the material can withstand before fracturing. Because the peak of curve A is higher on the stress axis than the peak of curve B, **Material A is also the stronger material**.

Practice Question 3 (Numerical — Ratio of Young's Moduli)

A steel wire of length $4.7\,\text{m}$ and cross-sectional area $3.0 \times 10^{-5}\,\text{m}^2$ stretches by the exact same amount as a copper wire of length $3.5\,\text{m}$ and cross-sectional area $4.0 \times 10^{-5}\,\text{m}^2$ under a given load. What is the ratio of the Young's modulus of steel to that of copper ($Y_{\text{steel}} : Y_{\text{copper}}$)?

(1) $1.79 : 1$

(2) $1.12 : 1$

(3) $0.56 : 1$

(4) $2.35 : 1$

Correct Answer: (1)

Worked Solution: Given that both wires carry the same load F and undergo the same elongation ΔL:

$$\Delta L = \frac{FL_s}{A_s Y_s} = \frac{FL_c}{A_c Y_c}$$

Cancelling F and rearranging for the ratio $\frac{Y_s}{Y_c}$:

$$\frac{Y_s}{Y_c} = \left(\frac{L_s}{L_c}\right) \times \left(\frac{A_c}{A_s}\right) = \left(\frac{4.7\,\text{m}}{3.5\,\text{m}}\right) \times \left(\frac{4.0 \times 10^{-5}\,\text{m}^2}{3.0 \times 10^{-5}\,\text{m}^2}\right) = \frac{18.8}{10.5} \approx 1.79$$

Thus, the ratio of the Young's modulus of steel to that of copper is $1.79 : 1$.

Practice Question 4 (Statement Analysis — Elasticity & Helical Springs)

Consider the following two statements regarding the mechanical properties of solids:

- **Statement I:** The Young's modulus of rubber is greater than that of steel because rubber stretches much more easily than steel for the same applied tension.
- **Statement II:** The stretching of a helical coil spring without permanent change in wire length is determined by its shear modulus.

In the light of the above statements, choose the correct option:

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true, but Statement II is false.

(4) Statement I is false, but Statement II is true.

Correct Answer: (4)

Worked Solution:

• **Statement I is false:** For a given load F and geometry (L, A), $Y = \frac{FL}{A\Delta L}$, so $Y \propto \frac{1}{\Delta L}$. Because rubber stretches much more ($\Delta L_{\text{rubber}} \gg \Delta L_{\text{steel}}$) under the same deforming force, the Young's modulus of rubber is much *smaller* than that of steel.

• **Statement II is true:** When a coil spring is stretched, neither the length of the wire forming the coil nor its volume changes appreciably; instead, the wire undergoes twisting (shearing deformation), which changes the overall shape of the coil. Hence, the stretching of a coil spring is determined by its **shear modulus (G)**.

Practice Question 5 (Numerical — Shear Deflection of a Cube)

The edge of an aluminium cube is $10\,\text{cm}$ long. One face of the cube is firmly fixed to a vertical wall. A mass of $100\,\text{kg}$ is then attached to the opposite face of the cube. If the shear modulus of aluminium is $25\,\text{GPa}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$, what is the vertical deflection of this face?

(1) $3.92 \times 10^{-7}\,\text{m}$

(2) $2.50 \times 10^{-6}\,\text{m}$

(3) $3.92 \times 10^{-5}\,\text{m}$

(4) $9.80 \times 10^{-7}\,\text{m}$

Correct Answer: (1)

Worked Solution:

Edge length of the cube: $L = 10\,\text{cm} = 0.1\,\text{m}$.

Area of the face subjected to the tangential force: $A = L^2 = (0.1\,\text{m})^2 = 0.01\,\text{m}^2$.

Tangential force due to the suspended mass: $F = mg = 100\,\text{kg} \times 9.8\,\text{m}\,\text{s}^{-2} = 980\,\text{N}$.

Shear modulus: $G = 25\,\text{GPa} = 25 \times 10^9\,\text{N}\,\text{m}^{-2}$.

From $G = \frac{FL}{A\Delta x}$, the vertical deflection Δx is:

$$\Delta x = \frac{FL}{AG} = \frac{980\,\text{N} \times 0.1\,\text{m}}{(0.01\,\text{m}^2) \times (25 \times 10^9\,\text{N}\,\text{m}^{-2})} = \frac{98}{2.5 \times 10^8}\,\text{m} = 3.92 \times 10^{-7}\,\text{m}$$

Practice Question 6 (Numerical — Hollow Cylindrical Columns)

Four identical hollow cylindrical columns of mild steel support a big structure of mass $50,000\,\text{kg}$. The inner and outer radii of each column are $30\,\text{cm}$ and $60\,\text{cm}$ respectively. Assuming the load distribution to be uniform, calculate the compressional strain of each column. (Given: $Y_{\text{steel}} = 2.0 \times 10^{11}\,\text{Pa}$, $g = 9.8\,\text{m}\,\text{s}^{-2}$.)

(1) 7.22×10^{-7}

(2) 2.89×10^{-6}

(3) 1.44×10^{-6}

(4) 3.61×10^{-7}

Correct Answer: (1)

Worked Solution:

Total weight of the structure: $W = Mg = 50,000\,\text{kg} \times 9.8\,\text{m}\,\text{s}^{-2} = 4.9 \times 10^5\,\text{N}$.

Since the load is shared equally by 4 columns, the compressive force on each column is:

$$F = \frac{4.9 \times 10^5\,\text{N}}{4} = 1.225 \times 10^5\,\text{N}$$

Cross-sectional area of each hollow cylinder ($r_1 = 0.30\,\text{m}$, $r_2 = 0.60\,\text{m}$):

$A = \pi(r_2^2 - r_1^2) = 3.14 \times (0.60^2 - 0.30^2)\,\text{m}^2 = 3.14 \times (0.36 - 0.09)\,\text{m}^2 = 3.14 \times 0.27\,\text{m}^2 \approx 0.8478\,\text{m}^2$

Compressional strain ε of each column:

$$\varepsilon = \frac{F}{AY} = \frac{1.225 \times 10^5\,\text{N}}{0.8478\,\text{m}^2 \times 2.0 \times 10^{11}\,\text{Pa}} \approx 7.22 \times 10^{-7}$$

Practice Question 7 (Numerical — Rigid Bar Supported by Three Parallel Wires)

A rigid horizontal bar of mass $15\,\text{kg}$ is supported symmetrically by three vertical wires, each $2.0\,\text{m}$ long. Those at each end are of copper and the middle one is of iron. If each wire is to have the exact same tension, what is the ratio of the diameter of each copper wire to the diameter of the iron wire ($d_{\text{copper}}/d_{\text{iron}}$)? (Take $Y_{\text{copper}} = 110 \times 10^9\,\text{Pa}$ and $Y_{\text{iron}} = 190 \times 10^9\,\text{Pa}$.)

(1) 1.31

(2) 1.73

(3) 0.76

(4) 0.58

Correct Answer: (1)

Worked Solution:

Because the bar is rigid and supported symmetrically, it remains horizontal, meaning all three wires of initial length $L = 2.0\,\text{m}$ undergo the exact same elongation ΔL, and hence have the exact same longitudinal strain $\varepsilon = \frac{\Delta L}{L}$.

From $Y = \frac{F/A}{\varepsilon} = \frac{4F}{\pi d^2 \varepsilon}$, since the tension F and strain ε are identical for all three wires:

$$Yd^2 = \text{constant} \implies \frac{d_{\text{copper}}}{d_{\text{iron}}} = \sqrt{\frac{Y_{\text{iron}}}{Y_{\text{copper}}}} = \sqrt{\frac{190 \times 10^9\,\text{Pa}}{110 \times 10^9\,\text{Pa}}} = \sqrt{\frac{19}{11}} = \sqrt{1.727} \approx 1.31$$

Practice Question 8 (Numerical — Bulk Modulus & Volume Contraction)

A solid copper cube, $10\,\text{cm}$ on an edge, is subjected to a uniform hydraulic pressure of $7.0 \times 10^6\,\text{Pa}$. If the bulk modulus of copper is $140\,\text{GPa}$, what is the volume contraction (ΔV) of the copper cube?

(1) $5.0 \times 10^{-2}\,\text{cm}^3$

(2) $5.0 \times 10^{-4}\,\text{cm}^3$

(3) $2.0 \times 10^{-2}\,\text{cm}^3$

(4) $7.0 \times 10^{-3}\,\text{cm}^3$

Correct Answer: (1)

Worked Solution:

Edge of the cube: $a = 10\,\text{cm}$. Original volume: $V = a^3 = (10\,\text{cm})^3 = 1000\,\text{cm}^3 = 10^{-3}\,\text{m}^3$.
Hydraulic pressure: $p = 7.0 \times 10^6\,\text{Pa}$. Bulk modulus: $B = 140\,\text{GPa} = 140 \times 10^9\,\text{Pa}$.
Magnitude of volume contraction ΔV:

$$\Delta V = \frac{pV}{B} = \frac{(7.0 \times 10^6\,\text{Pa}) \times (1000\,\text{cm}^3)}{140 \times 10^9\,\text{Pa}} = \frac{7.0 \times 10^9}{1.4 \times 10^{11}}\,\text{cm}^3 = 5.0 \times 10^{-2}\,\text{cm}^3$$

Practice Question 9 (Numerical — Elastic Potential Energy)

A piece of copper having a rectangular cross-section of $15.2\,\text{mm} \times 19.1\,\text{mm}$ is pulled in tension with a $44,500\,\text{N}$ force, producing only elastic deformation. Given that Young's modulus of copper is $1.1 \times 10^{11}\,\text{N}\,\text{m}^{-2}$, calculate (i) the resulting longitudinal strain and (ii) the elastic potential energy stored per unit volume of the copper piece.

(1) Strain $= 1.39 \times 10^{-3}$; Energy density $= 1.07 \times 10^{5}\,\text{J}\,\text{m}^{-3}$

(2) Strain $= 1.39 \times 10^{-3}$; Energy density $= 2.13 \times 10^{5}\,\text{J}\,\text{m}^{-3}$

(3) Strain $= 2.78 \times 10^{-3}$; Energy density $= 1.07 \times 10^{5}\,\text{J}\,\text{m}^{-3}$

(4) Strain $= 1.53 \times 10^{-4}$; Energy density $= 5.35 \times 10^{4}\,\text{J}\,\text{m}^{-3}$

Correct Answer: (1)

Worked Solution:

Cross-sectional area: $A = 15.2 \times 10^{-3}\,\text{m} \times 19.1 \times 10^{-3}\,\text{m} = 2.903 \times 10^{-4}\,\text{m}^2$.

Tensile stress: $\sigma = \frac{F}{A} = \frac{44{,}500\,\text{N}}{2.903 \times 10^{-4}\,\text{m}^2} = 1.533 \times 10^{8}\,\text{N}\,\text{m}^{-2}$.

(i) Resulting longitudinal strain (ε):

$$\varepsilon = \frac{\sigma}{Y} = \frac{1.533 \times 10^{8}\,\text{N}\,\text{m}^{-2}}{1.1 \times 10^{11}\,\text{N}\,\text{m}^{-2}} \approx 1.39 \times 10^{-3}$$

(ii) Elastic potential energy per unit volume (u):

$$u = \frac{1}{2}\sigma\varepsilon = \frac{1}{2} \times (1.533 \times 10^{8}\,\text{N}\,\text{m}^{-2}) \times (1.39 \times 10^{-3}) \approx 1.07 \times 10^{5}\,\text{J}\,\text{m}^{-3}$$

Practice Question 10 (Numerical — Poisson's Ratio & Lateral Contraction)

A cylindrical steel wire of length $2.0\,\text{m}$ and diameter $4.0\,\text{mm}$ is stretched within its elastic limit such that its length increases by $2.0\,\text{mm}$. If Poisson's ratio for this steel is 0.29, by how much does the diameter of the wire contract?

(1) $1.16\,\mu\text{m}$

(2) $2.90\,\mu\text{m}$

(3) $0.58\,\mu\text{m}$

(4) $4.00\,\mu\text{m}$

Correct Answer: (1)

Worked Solution:

Longitudinal strain: $\frac{\Delta L}{L} = \frac{2.0 \times 10^{-3}\,\text{m}}{2.0\,\text{m}} = 1.0 \times 10^{-3}$.

From the definition of Poisson's ratio ($\frac{\Delta d/d}{\Delta L/L} = 0.29$), the lateral strain is:

$$\frac{\Delta d}{d} = 0.29 \times \left(\frac{\Delta L}{L}\right) = 0.29 \times 10^{-3}$$

Given original diameter $d = 4.0\,\text{mm} = 4.0 \times 10^{-3}\,\text{m}$, the contraction in diameter Δd is:

$$\Delta d = (0.29 \times 10^{-3}) \times (4.0 \times 10^{-3}\,\text{m}) = 1.16 \times 10^{-6}\,\text{m} = 1.16\,\mu\text{m}$$

Practice Question 11 (Assertion–Reason — Beam Design)

Assertion (A): Load-bearing beams used in the construction of bridges and buildings are commonly manufactured with an I-shaped cross-section rather than a simple rectangular cross-section.

Reason (R): Increasing the depth d of a beam reduces central sagging significantly ($\delta \propto d^{-3}$), and the wide top and bottom flanges of an I-section prevent lateral buckling while reducing the weight and cost of the beam.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

Correct Answer: (1)

Worked Solution: From $\delta = \frac{Wl^3}{4bd^3Y}$, sagging is inversely proportional to d^3. However, a simple deep thin bar buckles sideways when the load is not perfectly centred. An I-shaped section combines large depth d (to minimize sagging) with wide horizontal flanges (to provide a large load-bearing surface and resist buckling) and a thin vertical web (to save weight and cost). Thus, both Assertion and Reason are true, and Reason is the exact explanation of Assertion.

Practice Question 12 (Numerical — Chairlift Cable & Water Compression)

A steel cable with a radius of $1.5\ \text{cm}$ supports a chairlift at a ski area. If the maximum stress is not to exceed $10^8\ \text{N m}^{-2}$, what is the maximum load the cable can support? Also, how much should the pressure on $1\ \text{litre}$ of water ($B = 2.2 \times 10^9\ \text{Pa}$) be changed to compress it by 0.10%?

(1) Maximum load $= 7.07 \times 10^4\ \text{N}$; Pressure change $= 2.2 \times 10^6\ \text{Pa}$

(2) Maximum load $= 3.53 \times 10^4\ \text{N}$; Pressure change $= 2.2 \times 10^6\ \text{Pa}$

(3) Maximum load $= 7.07 \times 10^4\ \text{N}$; Pressure change $= 2.2 \times 10^5\ \text{Pa}$

(4) Maximum load $= 1.50 \times 10^5\ \text{N}$; Pressure change $= 4.4 \times 10^6\ \text{Pa}$

Correct Answer: (1)

Worked Solution:

1. Maximum Load on the Cable:

Radius $r = 1.5\ \text{cm} = 1.5 \times 10^{-2}\ \text{m}$. Cross-sectional area
$A = \pi r^2 = 3.1416 \times (1.5 \times 10^{-2}\ \text{m})^2 = 7.068 \times 10^{-4}\ \text{m}^2$.
Maximum load: $F_{\max} = \sigma_{\max} \times A = (10^8\ \text{N m}^{-2}) \times (7.068 \times 10^{-4}\ \text{m}^2) \approx 7.07 \times 10^4\ \text{N}$.

2. Pressure Change to Compress Water by 0.10%:

Given fractional volume compression $\frac{\Delta V}{V} = 0.10\% = \frac{0.10}{100} = 1.0 \times 10^{-3}$:

$$\Delta p = B\left(\frac{\Delta V}{V}\right) = (2.2 \times 10^9\ \text{Pa}) \times (1.0 \times 10^{-3}) = 2.2 \times 10^6\ \text{Pa}$$

7. Common Pitfalls & Quick Revision Checklist

Common Pitfalls to Avoid in NEET UG

1. **The "More Elastic" Misconception:** In everyday language, people think a material that stretches more (like rubber) is "more elastic." In physics, this is a misnomer! A material that stretches to a *lesser* extent for a given load has a larger Young's modulus and is scientifically **more elastic** (hence steel is more elastic than rubber, copper, brass, and aluminium).
2. **Tension in a Suspended Wire:** When a wire hangs from a ceiling supporting a load F, the ceiling pulls upward with force F and the weight pulls downward with force F. Never double the force to $2F$! The internal tension at any cross-section is simply F, so stress is F/A.
3. **Applicability of Elastic Moduli to States of Matter:** Young's modulus (Y) and Shear modulus (G) are relevant **only for solids** because only solids possess definite lengths and shapes. Bulk modulus (B) is relevant for **all three states of matter** (solids, liquids, and gases).
4. **Vector vs. Tensor Nature of Stress:** Unlike force, **stress is not a vector quantity** because it cannot be assigned a single specific direction without specifying the orientation of the surface element on which the internal forces act.
5. **Factor of $\frac{1}{2}$ in Elastic Potential Energy:** While the work done by the external stretching force can be related to the final force F and final elongation l, the average restoring force during stretching from 0 to l is $\frac{1}{2}F$. Thus, the stored elastic potential energy is $\frac{1}{2}Fl$ (and energy density is $\frac{1}{2}\sigma\varepsilon$).

Concise Final Formula Recap

- **Longitudinal Stress & Strain:** $\sigma = \frac{F}{A}, \varepsilon = \frac{\Delta L}{L}, Y = \frac{\sigma}{\varepsilon} = \frac{FL}{A\Delta L}$
- **Shearing Stress & Strain:** $\sigma_s = \frac{F}{A}, \theta \approx \tan\theta = \frac{\Delta x}{L}, G = \frac{\sigma_s}{\theta} = \frac{FL}{A\Delta x}$ (For most solids, $G \approx \frac{Y}{3}$)
- **Hydraulic Stress, Volume Strain & Bulk Modulus:** $B = -\frac{p}{\Delta V/V}$, Compressibility $k = \frac{1}{B} = -\frac{1}{\Delta p}\left(\frac{\Delta V}{V}\right)$
- **Poisson's Ratio:** $\frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{\Delta d/d}{\Delta L/L}$
- **Elastic Potential Energy:** $U = \frac{1}{2}F\Delta L = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$; Energy density $u = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}Y\varepsilon^2$
- **Central Sagging of a Beam:** $\delta = \frac{Wl^3}{4bd^3Y}$
- **Maximum Height of a Mountain:** $h_{\max} = \frac{\sigma_{\text{elastic limit, shear}}}{\rho g} \approx 10\,\text{km}$